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Studio Session	01

Part A: URL of the Visualisation

https://zche0436.github.io/FIT2179-DataVisualisation2/

Part B: Domain Description

Describe your domain chosen for DV II

The domain of this visualisation is monsoon flooding and flood related economic loss in Malaysia. It explores how seasonal rainfall and major flood events create uneven impacts across Malaysian states, affecting homes, agriculture, public infrastructure, businesses, vehicles and manufacturing. The visualisation focuses on the relationship between rainfall patterns, geographic flood exposure and the financial consequences of flooding. This helps users understand where losses are concentrated and which sectors are most vulnerable.

Describe the Why and Who aspects of your DV II

Who: The primary audience is the average Malaysian public, especially residents in flood-prone areas, households, students and everyday news readers who want to understand the real impact of monsoon flooding beyond rainfall headlines. The secondary audience includes local authorities, disaster response planners and community organisations that need a clearer view of which states and sectors experience the greatest flood-related losses.

Why: The overarching goal is to present a compelling and data-driven narrative that reveals how monsoon flooding creates uneven economic and social impacts across Malaysia. The visualisation connects rainfall patterns, state-level flood exposure and sector-based losses to help users understand that flood damage is not evenly distributed but shaped by geography, land use, infrastructure and local vulnerability. The agricultural section is included because crop and farming losses show how flooding affects not only property and infrastructure but also food production, rural livelihoods and the wider economy.

- Comparing rainfall intensity across months and years
- Identifying which Malaysian states experience the highest flood related losses
- Discovering which sectors carry the greatest financial burden
- Evaluating the vulnerability of agriculture
- Revealing changes in damage ranking over time

Describe the "What" aspect of your DV II

Source: The rainfall data is sourced from the World Bank Climate Change Knowledge Portal which provides climate and precipitation data for Malaysia. The flood loss and agricultural loss data are sourced from the Department of Statistics Malaysia (DOSM) through official flood damage statistics.

Relevance: These datasets are relevant because they connect climate conditions with real flood consequences. The rainfall data provides the environmental context behind Malaysia's monsoon flooding while DOSM's flood loss data shows the economic impact across states and sectors. Together, they help reveal not only where flooding occurs but also which assets, sectors and communities are most affected.

Creation Process: The raw datasets were cleaned, filtered and reshaped into formats suitable for visualisation. Rainfall data from the World Bank Climate Change Knowledge Portal was aggregated by month and year to reveal seasonal monsoon patterns. DOSM flood loss data was grouped by state and damage category, allowing comparison of losses across locations and sectors. Agricultural loss data was also reorganised by state, crop type and loss value that enable more detailed analysis of which crops and regions were most affected by flooding.

Describe the "How" aspect of your DV II

The visualisation uses a partitioned scrollytelling dashboard structure to guide users through Malaysia's flood story from climate trigger to economic consequences. Each section focuses on a different analytical task such as understanding rainfall patterns, comparing state-level losses, identifying vulnerable sectors and exploring agricultural damage. This creates an author-driven narrative while still allowing users to interact with filters, maps and charts based on their own curiosity.

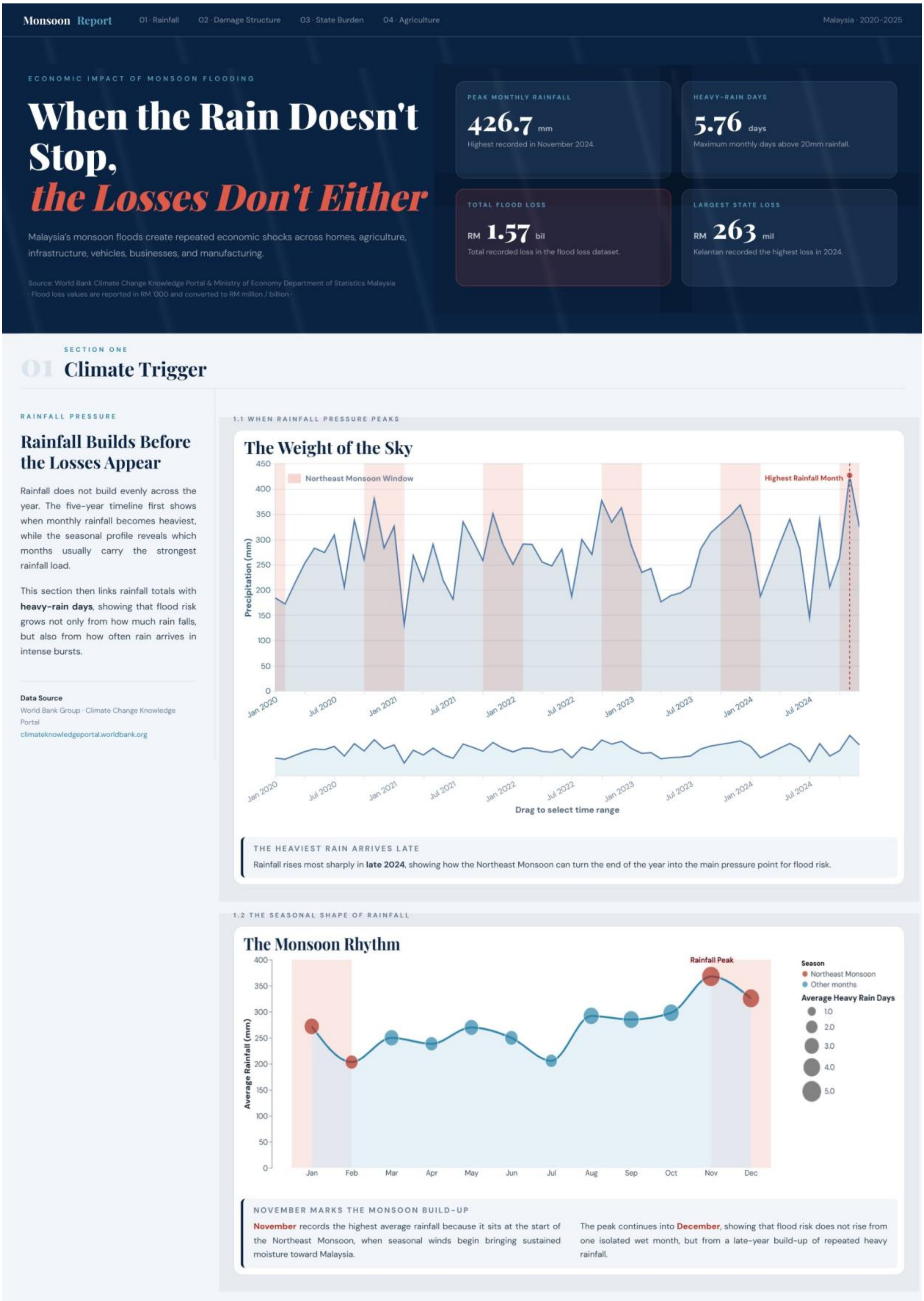
- Rainfall Area Chart
  The area chart shows rainfall intensity over time, making seasonal build-up and repeated monsoon peaks easy to identify. Its continuous shape helps users view rainfall as an accumulating climate pressure rather than isolated monthly values.
- Scatter Plot
  This plot shows the relationship between monthly rainfall and the number of heavy rain days. It uses x and y position channels for relationship-based analysis. The trend line helps users identify the positive relationship between rainfall volume and repeated heavy rain days while colour hue distinguishes Northeast Monsoon months from other months. This helps users understand how rainfall bursts build flood pressure before moving into the loss analysis.
- Choropleth Map
  The map uses Malaysian states as area marks and colour luminance to encode each state's percentage contribution to national flood loss. Instead of showing only raw loss values, the data is normalised as a percentage of total national loss, making it easier to compare the relative burden of each state. This helps users quickly identify which states contribute most of Malaysia's overall flood damage and supports a more meaningful spatial comparison across regions. Darker shades represent a higher share of national flood damage while lighter shades represent a lower share.
- Donut Chart
  The donut chart shows how total flood loss is divided across categories in a state selected from the lollipop chart. This helps users understand not only where losses occur but also what types of damage contribute most to the overall impact.
- Agricultural Loss Charts
  These charts break down flood damage by state and crop type, helping users identify which agricultural areas and commodities are most vulnerable. This provides a more detailed perspective because agricultural losses may be overlooked when they are combined with broader flood damage categories. By separating losses into crop categories, users can better understand how floods affect farming communities, rural livelihoods and food production across different Malaysian states.

- Interactive Features:
  The visualisation includes interactive elements such as drop down box selection, small multiples linked charts, hover tooltips and overview-to-detail brush selection. The rainfall chart allows users to drag across the overview timeline to select a specific period which updates the detailed chart above and supports focused time-based exploration. Small multiples allow repeated charts to be compared side by side using the same scale and structure, making patterns across states or categories easier to identify. Tooltips provide exact values without overcrowding the charts while linked interactions help users move from national-level patterns to more detailed state, sector and crop-level insights.

Part C: Sketch

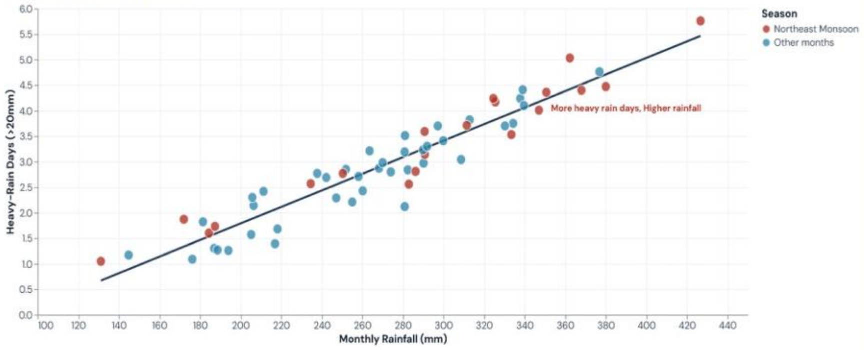
https://github.com/zche0436/FIT2179-DataVisualisation2/blob/main/DV2-Sketch.pdf

Part D: Full-size screenshot of your DV II



1.3 RAINFALL BURST PATTERN

Bursts Before Floods



REPEATED BURSTS RAISE FLOOD PRESSURE

The upward trend shows that months with higher rainfall totals usually also record **more heavy-rain days**. Flood pressure builds when rain arrives repeatedly in intense bursts, not only from one large monthly total.

Monsoon months appear more often on the higher-rainfall side of the chart, linking the seasonal rainfall surge to stronger short-term flood intensity.

SECTION TWO

02 The Anatomy of Flood Damage

DAMAGE STRUCTURE

Behind Every Flooded State, the Damage Pattern Is Different

This section breaks flood damage into its internal structure. It shows which **sectors** drive losses, where damage is concentrated within key **districts**, and how severe losses become when measured against exposed **living quarters**.

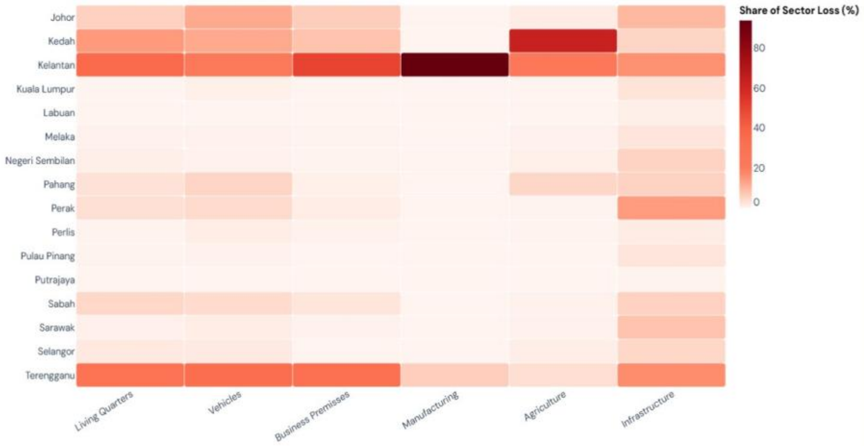
Data Source
Ministry of Economy Department of Statistics Malaysia
www.dosm.gov.my

2.1 STATE X SECTOR LOSS MATRIX

READING THE DAMAGE PATTERN

Darker red cells show larger flood losses. Compare each row to see which sector affects a state most and compare each column to see where a sector causes the greatest damage.

Where each Sector Hurts Most



Select Year: 2024

MANUFACTURING LOSS HIGHLY CONCENTRATED

Kelantan carried **93.8%** of manufacturing flood loss in **2024**, while **Terengganu** carried **100%** in **2025**.

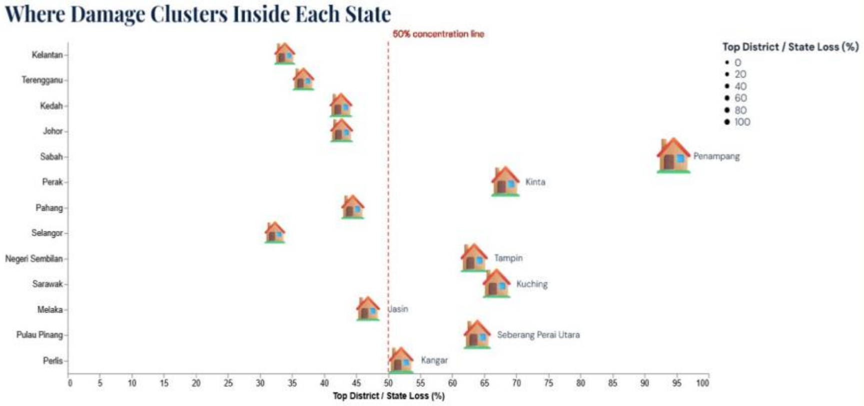
As Kelantan and Terengganu sit along Malaysia's **east coast flood belt** where Northeast Monsoon flooding often causes severe river overflow.
This makes manufacturing sites, transport routes and local supply chains more exposed when floodwaters spread beyond residential areas.

2.2 DISTRICT-LEVEL DAMAGE CONCENTRATION

READING THE DISTRICT CLUSTER

Larger symbols and positions further right both show stronger district concentration. Hover to see the actual flood loss value in RM '000.

Where Damage Clusters Inside Each State



Select Year: 2024

FLOOD DAMAGE CAN CLUSTER LOCALLY

The repeated house symbol shows that the top-loss districts are mainly dominated by **Living Quarters** damage. A district can record a large loss value but still have a lower share if the state's flood damage is spread across many districts.

A smaller loss can show a higher percentage when most of the state's flood damage is concentrated in one district.

2.3 LOSS PRESSURE BY RESIDENTIAL EXPOSURE

Flood Loss Pressure Beyond Raw Totals

EXPOSURE CHANGES THE RANKING OF DAMAGE

Longer arcs show higher flood loss per **1,000 living quarters**, revealing where damage is heavier relative to residential exposure.

This view reduces the bias of raw totals, where larger states may appear more severe simply because they contain more homes and assets.

SECTION THREE

03 The Geography of Loss

STATE-LEVEL BURDEN

Some States Carry More of the National Flood Burden

Flood damage concentrates in specific parts of Malaysia rather than spreading evenly across the country. In 2024, the top three states account for more than two-thirds of national flood loss. Selecting a state reveals the category mix behind its total loss, linking state ranking with damage composition.

Data Source
Ministry of Economy Department of Statistics

3.1 STATE SHARE OF NATIONAL FLOOD LOSS (%)

Malaysia
www.dosm.gov.my

Where the Water Leaves Its Mark

Select Year: 2024

LOSS IS CONCENTRATED IN A FEW STATES

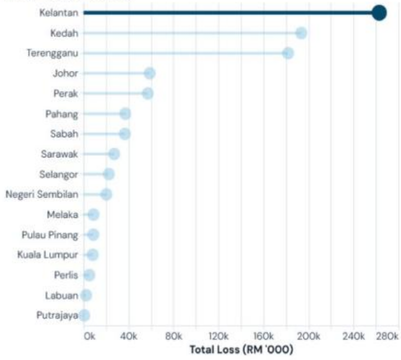
In **2024**, flood loss is strongly concentrated in **Kelantan**, **Kedah**, and **Terengganu**. Together, these three states account for about **68.4%** of the national recorded flood loss.

The map shows that national flood impact is not evenly spread. The darkest states form the main loss belt, meaning a small number of states carry most of the flood damage burden.

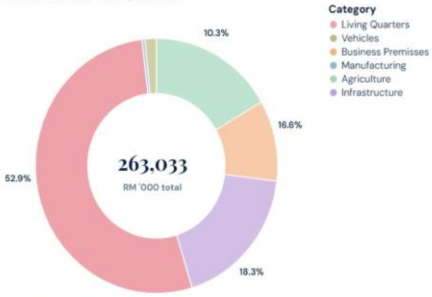
3.2 STATE RANKING & LOSS COMPOSITION

The States that Carry the Cost

State Loss Ranking



Kelantan Loss Composition



Main Damage Driver:
Living Quarters makes up 52.9% of Kelantan flood losses

Select Year: 2024

HIGH TOTALS HIDE DIFFERENT DAMAGE MIXES

In **2024**, **Kelantan** records the highest total loss, but its damage is not from one category only. Living quarters, infrastructure, agriculture, and business premises all contribute to the total.

The linked donut helps explain why state rankings alone are not enough. For example, **Kedah** ranks high mainly because of agricultural loss, while **Terengganu** is driven more by living quarters and infrastructure.

SECTION FOUR

04 Agricultural Flood Losses

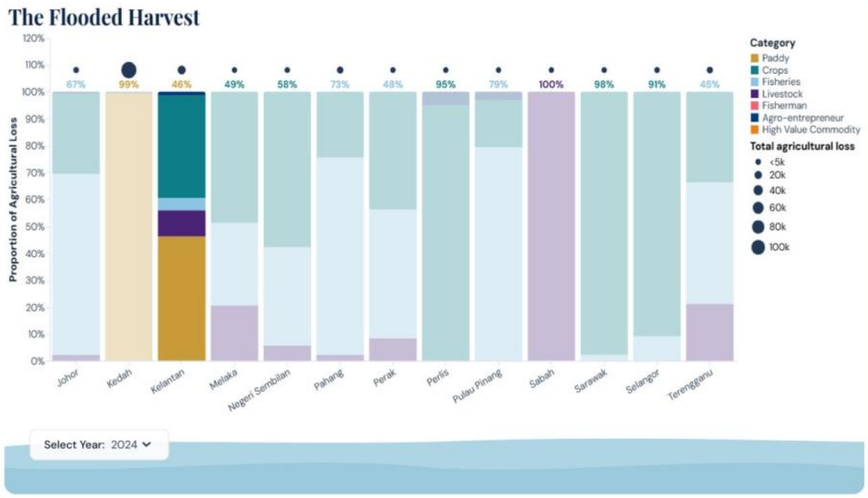
AGRICULTURAL DEEP DIVE

The Harvest Lost to Floodwaters

Agricultural flood loss is first compared across all states, then narrowed into a selected-state view. The stacked chart shows how each state's agricultural damage is divided by category, while the detail chart reveals the actual loss value and whether each category rose or fell from 2024 to 2025.

Data Source
Ministry of Economy Department of Statistics Malaysia
www.dosm.gov.my

4.1 AGRICULTURAL LOSS COMPOSITION ACROSS STATES

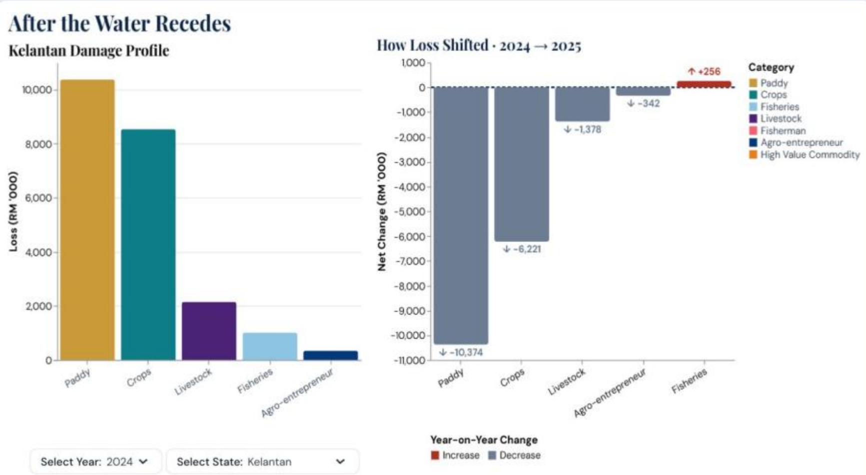


PADDY LOSS DOMINATES THE AGRICULTURAL PICTURE

In 2024, agricultural flood loss is dominated by Kedah, where paddy loss makes up most of the state's agricultural damage. This makes the 2024 agricultural pattern highly concentrated.

In 2025, the pattern becomes more distributed across states such as Johor, Selangor, Kedah, and Kelantan. Fisheries and crops become more visible, while paddy loss no longer dominates the chart.

4.2 SELECTED-STATE DAMAGE PROFILE & YEAR-ON-YEAR SHIFT



READING THE SHIFT

In Kelantan, the largest 2024 agricultural loss came from Paddy.
From 2024 to 2025, the biggest shift was Paddy,
which decreased by RM 10,374'000.
This means its 2025 loss was lower than the previous year.

The dashed zero line in How Loss Shifted marks no change from 2024 to 2025. Bars above the line indicate higher losses, while bars below the line indicate reduced losses.

METADATA

Author: Chen Zhe Tein (36633909)
Project: FIT2179 Data Visualisation
Topic: Malaysia Monsoon Flood Impact
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DATA SOURCES

DOSM Special Report on Impact of Floods in Malaysia 2025
DOSM Special Report on Impact of Floods in Malaysia 2024
World Bank Climate Change Knowledge Portal
Rainfall data covers 2020-2024. Flood loss data covers 2024-2025.